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MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The design requires enough stability so that if it is knocked off of the desk, it won't shatter. This is because the product will be used by middle schoolers, who may accidentally knock it over.

Design Requirement #2 (HIGH Priority)

The product requires a design that's eye-catching enough to easily create opportunities for conversation about it, letting kids gain an easy opportunity to socialize. This a

Design Requirement #3 (HIGH Priority)

The maximum measurements for this product will be 10 by 10 inches. This is high priority, because the librarian told us that it wouldn't be able to fit onto the desk if it was any larger.

Design Requirement #4 (HIGH Priority)

The whiteboard on which the librarian would write the questions would need to be easily accessible, removable, and erasable. This would create a convient and cost-efficient way for the librarian to swap out the question, allowing the machine to have a certain question for a set amount of time, ex. a month, before being swapped out for new options.

Design Requirement #5 (HIGH Priority)

The jars that hold the pom-poms should be unscrewable.

This means that the librarian can very easily measure the responses, without having to guess the amount of pom-poms.

Design Requirement #6 (LOW Priority)

The design should not be too expensive, so that ideally the product can be replicated.

If the product works, then it follows logically that it should be used at a larger scale, so that more kids can have more access to social engagement and have less risk of burnout.

Design Requirement #7 (LOW Priority)

Lastly, the product should be easy to build at home, and not require expensive or unattainable materials.

This means that not only would the product be far easier for us as engineers to design, but for the librarian to understand, and for others to create replicas.

The librarian would have a lot easier time using the machine if it didn’t require complex parts or moving machinery.